MLOps Exam Assignment

Weather Rain Prediction MLOps Pipeline

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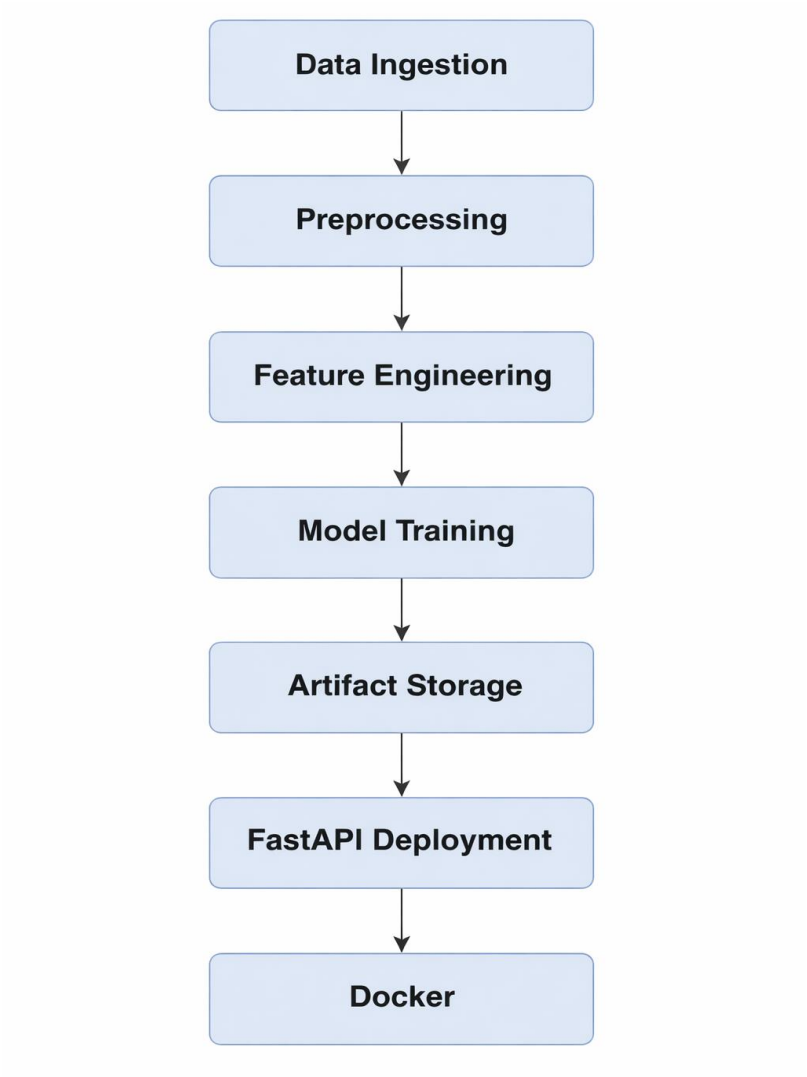
1. Introduction

This project aims to develop a complete MLOps pipeline for weather prediction. The system predicts whether it will rain tomorrow based on historical weather data.
The focus of the project is not only model performance but also building a reproducible and deployable machine learning pipeline. The system includes data ingestion, preprocessing, model training, deployment via API, and monitoring.
The solution is designed to simulate a real-world production environment.

2. Pipeline Overview

The pipeline consists of the following components:

- Data Ingestion: Weather data is collected from the Open-Meteo API.
- Preprocessing: Data is cleaned and transformed, and a target variable is created.
- Feature Engineering: Relevant features such as temperature and precipitation are used.
- Model Training: A Logistic Regression model is trained.
- Deployment: The model is deployed using FastAPI.
- Monitoring: Predictions are logged for tracking.
- Containerization: The system is packaged using Docker.



MLOps Pipeline for Weather Prediction System

This diagram illustrates the complete workflow of the MLOps pipeline from data ingestion to deployment and monitoring.

3. Data Ingestion

The pipeline supports both original and new data ingestion. The first execution collects the initial weather dataset from the Open-Meteo API (original data). When the pipeline is run again, it fetches updated weather data (new data).

The ingestion process is automated using a Python script, which retrieves the data and stores it locally in CSV format. This ensures reproducibility and allows the pipeline to be executed repeatedly with updated data.

The raw data is stored in the data/raw directory.

4. Data Preprocessing

The raw data is processed to prepare it for model training. A new target variable called **rain_tomorrow** is created based on precipitation values.

The precipitation values are shifted by one day to represent whether it will rain the next day. The target is then converted into a binary classification problem, where 1 indicates rain and 0 indicates no rain.

Missing values created during shifting are removed to ensure clean data.

5. Feature Engineering

The model uses the following features:

- Maximum temperature
- Minimum temperature
- Precipitation

These features are selected based on their relevance to weather conditions and their influence on rainfall prediction.

The feature selection is kept simple to focus on building a robust and deployable MLOps pipeline.

6. Model Training

A Logistic Regression model is used for prediction. The dataset is split into training and testing sets using an 80/20 ratio.

The model is trained on the training dataset and evaluated on the test dataset. The performance is measured using accuracy and classification metrics such as precision, recall, and F1-score.

The trained model is saved as a serialized file for later use in deployment.

7. Results

The model was evaluated on the test dataset after training. The achieved accuracy was approximately **0.61**.

The classification report shows that the model can predict both classes (rain and no rain), although performance is limited due to the relatively small dataset.

```
✅ Model trained successfully
Accuracy: 0.6111

Classification Report:

              precision    recall  f1-score   support

           0       0.62      0.80      0.70        10
           1       0.60      0.38      0.46         8

    accuracy                           0.61        18
   macro avg       0.61      0.59      0.58        18
weighted avg       0.61      0.61      0.59        18

✅ Model saved to models/rain_model.pkl
```

Example API prediction:

Input:

```
{
  "temp_max": 20,
  "temp_min": 10,
  "precipitation": 2
}
```

Output:

```
{
  "prediction": 0,
  "result": "No rain tomorrow"
}
```

8. Deployment

The system is triggerable on demand through the FastAPI /predict endpoint. A user can send input data through an HTTP request and receive a prediction immediately. This satisfies the requirement that the pipeline should be runnable continuously or triggered when needed.

This allows the model to be easily integrated into other systems and simulates a real-world production environment.

9. Monitoring

Monitoring is implemented through logging in the API. Each prediction request is stored in a log file together with the input values and prediction result. This makes it possible to track how the model is used and inspect its behaviour over time.

10. Reproducibility and Artifact Management

The pipeline stores important artifacts in structured folders. Raw weather data is saved in data/raw/weather.csv, processed data is stored in the processed data folder, the trained model is stored in models/rain_model.pkl, and API prediction logs are stored in logs.log.

11. Containerization

The application is containerized using Docker to ensure consistent execution across different environments.

A Docker file is created to define the environment, install dependencies, and run the API server.

12. Conclusion

This project demonstrates the development of a complete MLOps pipeline for weather prediction.

The system integrates data ingestion, preprocessing, model training, deployment, and monitoring into a single workflow.

The focus on reproducibility and deployment ensures that the solution can be used in real-world scenarios.